# Session 1.2 Activity

AI can already do four of these five things, at least in part. The one it cannot do yet is a full surgery on a real person.

| Task | Can AI do it? | In short |
|---|---|---|
| **1. Playing table tennis** | **YES** | A robot beat top human players |
| **2. Discovering new math** | **YES (some problems)** | Solves real unsolved puzzles, not everything |
| **3. Writing a funny story** | **MOSTLY** | Good at short jokes, less proven for long stories |
| **4. English speech to French speech** | **YES** | Works live with about a 2 second delay |
| **5. Complex surgery** | **NOT YET** | Works on lab tissue, not on real patients |

## 1. Playing table tennis

**Yes.** In 2026, Sony built a robot arm called Ace that plays table tennis by the official rules. It beat several top players, including professionals. It still lost some matches, so it is not perfect.

**Keep in mind:** It is one robot arm in a special lab, not a robot that can play any sport.

## 2. Discovering new math

**Yes, for some problems.** Google DeepMind built a system that solved 9 famous unsolved math problems, and two of them had been open for 56 years. A computer program then checked every step of the proofs, so they are not just guesses.

**Keep in mind:** It tried 353 problems and solved 9. That is a real achievement, but AI is not a general math genius yet. Some results are also still waiting for expert review.

## 3. Writing a funny story

**Mostly yes.** In tests, people rated AI jokes as funny as jokes written by humans. In one test they were close to the professional writers at The Onion. Newer AI models also scored near human writers on humor in short stories.

**Keep in mind:** Most tests used short jokes. Nobody has really tested whether AI can keep a long story funny from start to finish. Humor is also personal, so what is funny to one person may not be funny to another.

## 4. Translating spoken English into spoken French

**Yes.** Google has a system that listens to you speak and says the translation out loud, in a voice that sounds like yours. The delay is only about 2 seconds. English to French is one of the languages it supports.

**Keep in mind:** Faster translation means a bit less accuracy. Slang, strong accents, and noisy places can still cause mistakes.

## 5. Performing a complex surgery

**Not yet.** Researchers at Johns Hopkins taught a surgical robot to do a key part of gallbladder removal by itself. It did the job 8 times out of 8 with no human help.

**Keep in mind:** Those were pig organs outside the body, not a living person. A real patient bleeds, breathes, and moves, which makes it much harder. It also did one part of the surgery, not the whole thing. Doctors still need to prove it is safe before it can be used on people.

## The big picture

- AI is strongest when the result is easy to check (a point won, a proof verified) or when the job is digital (text and speech).
- AI is weakest when mistakes are dangerous and the real world is messy, like surgery.
- Some of these results come from the companies that built the systems, so treat them as promising, not final.